

Training Day 3 Report

Date: 19 June 2025

Module Name: Ethical Hacking Methodologies and Network Fundamentals

Mode: Online

Main Focus of the Module

Day 3 of the training focused on the structured methodologies used in ethical hacking, concepts of risk and threat management, and incident response. The second half of the session introduced **network fundamentals**, including IP addressing, subnetting, and calculating network ranges.

Topics Covered

Ethical Hacking Methodologies

- **Reconnaissance → Scanning → Gaining Access → Maintaining Access → Covering Tracks**
 - Each phase includes tools and strategies used by attackers and ethical hackers alike.
 - Helps understand where to detect and stop attacks during their lifecycle.
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☐ Defense in Depth (With Diagram)

Definition: A multi-layered approach to cybersecurity where several defense mechanisms are implemented to protect information.

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[Perimeter]	→ Firewalls, Gateways
[Network]	→ IDS/IPS, VLANs
[Endpoint]	→ Antivirus, Hardening
[Application]	→ Secure Coding, Patching
[Data]	→ Encryption, Access Control
[User Layer]	→ Training, Strong Authentication

Goal: Even if one layer fails, others continue to protect the system.

Risk, Threats & Vulnerabilities

- **Risk:** Probability that a threat will exploit a vulnerability
 - *Example:* Risk = Threat × Vulnerability × Impact

- **Threat:** Any potential danger (e.g., malware, attacker, insider)
 - **Vulnerability:** Weakness (e.g., unpatched system, misconfiguration)
 - **Risk Management:** Identify → Assess → Mitigate → Monitor risks
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Threat Modeling

- Process of identifying potential threats, vulnerabilities, and countermeasures.
 - Common methods: STRIDE, PASTA, DREAD
 - Helps prioritize what to secure in an application/network.
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Incident Management vs Incident Response

- **Incident Management:** Organizational approach to managing cyber incidents
 - **Incident Handling:** Step-by-step detection, analysis, containment, eradication
 - **Incident Response Plan (IRP):**
 - Preparation
 - Detection & Analysis
 - Containment
 - Eradication
 - Recovery
 - Lessons Learned
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Network Concepts: IP Addressing & Subnetting

IP Address Classes (Classful)

Class	Range	Default Subnet Mask	Usage
A	1.0.0.0 – 126.255.255.255	255.0.0.0	Very large networks
B	128.0.0.0 – 191.255.255.255	255.255.0.0	Medium networks
C	192.0.0.0 – 223.255.255.255	255.255.255.0	Small networks
D	224.0.0.0 – 239.255.255.255	N/A (Multicast)	Special use
E	240.0.0.0 – 255.255.255.255	Reserved	Research

☐ Identifying Network Bits, Host Bits & Subnet Mask

- **Example:** IP = 192.168.1.0/24

- Class C → Default subnet mask = 255.255.255.0
 - **Network bits:** 24
 - **Host bits:** 8
 - **No. of Hosts:** $2^8 - 2 = 254$
 - **First IP:** 192.168.1.1
 - **Last IP:** 192.168.1.254
 - **Broadcast IP:** 192.168.1.255
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🔗 Valid IP Range & Subnet Example

IP Address: 10.0.0.0/8

- **Subnet Mask:** 255.0.0.0
 - **Total Networks (with subnetting /16):** $2^8 = 256$
 - **Hosts per Subnet (for /16):** $2^{16} - 2 = 65,534$
 - **Valid IP Range (for subnet 10.0.0.0/16):**
 - **First Host IP:** 10.0.0.1
 - **Last Host IP:** 10.0.255.254
 - **Broadcast:** 10.0.255.255
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🧠 Key Takeaways

- Ethical hacking follows a structured approach similar to a kill chain
- Risk = Threat × Vulnerability; thus, eliminating either reduces risk
- Subnetting is crucial for logical network design and segmentation
- Understanding IP addressing helps identify network layouts and potential target